11 - Functions

Week 12 – 4 Nov 2025

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Cheat sheet

One-to-one functions.

Definition

Let F be a function from a set X to a set Y. F is **one-to-one** (or **injective**) if and only if for all elements x_1 and x_2 in X,

$$\text{if } F(x_1) = F(x_2), \text{ then } x_1 = x_2.$$

Or, equivalently,

$$\text{if } x_1 \neq x_2, \text{ then } F(x_1) \neq F(x_2).$$

Symbolically,

$$F : X \to Y \text{ is one-to-one} \iff \forall x_1, x_2 \in X, \text{ if } F(x_1) = F(x_2), \text{ then } x_1 = x_2.$$

$$F : X \to Y \textit{ is not one-to-one} \iff \exists x_1, x_2 \in X \textit{ with } F(x_1) = F(x_2) \textit{ but } x_1 \neq x_2.$$

Onto functions

Definition

Let F be a function from a set X to a set Y.

Then F is **onto** (or **surjective**) if and only if given any element y in Y, it is possible to find an element x in X with the property that $y = F(x)$.

Symbolically,

$$F : X \to Y \text{ is onto} \iff \forall y \in Y, \exists x \in X \text{ such that } F(x) = y.$$

$$F : X \to Y \textit{ is not onto} \iff \exists y \in Y \textit{ such that } \forall x \in X, F(x) \neq y.$$

F is both injective and surjective iff F is **bijective.**

Ex. 7.1.42, 43, 48. Let X and Y be sets, let A and B be any subsets of X and let C and D be any subsets of Y. Determine whether the properties are true for all functions F from X to Y and whether the properties are false for at least one function F from X to Y. Justify your answer.

42. $F(A \cap B) \subseteq F(A) \cap F(B)$.

43. $F(A) \cap F(B) \subseteq F(A \cap B)$.

48. For all subsets C and D of Y,

$$F^{-1}(C - D) = F^{-1}(C) - F^{-1}(D).$$

a. Define $H : \mathbb{R} \to \mathbb{R}$ by the rule $H(x) = x^2$, for all real numbers x.

(i) Is H one-to-one? Prove or give a counterexample.

(ii) Is H onto? Prove or give a counterexample.

SOLUTION . a. (i) No, H is not one-to-one. E.g., take $x_1 = 1$ and $x_2 = -1$. Then $H(1) = 1^2 = 1$ and $H(-1) = (-1)^2 = 1$. Therefore $H(1) = H(-1)$ but $1 \neq -1$.

(ii) No, H is not onto. Consider $y = -1$ in the co-domain $\mathbb{R}$. Suppose $y = -1$ has a pre-image, say x, then $H(x) = x^2 = -1$. Hence $x^2 = -1$ for some real number x, which is impossible. Hence $y = -1$ has no pre-image, and so H is not onto.

b. Define $K : \mathbb{R}^{\text{nonneg}} \to \mathbb{R}^{\text{nonneg}}$ by the rule $K(x) = x^2$, for all nonnegative real numbers x. Is K onto? Prove or give a counterexample.

b. K is onto.

Proof:

Suppose y is any element of $\mathbb{R}^{\text{nonneg}}$. Let $x = \sqrt{y}$. Then x is a real number because $y \geq 0$, and by definition of K,

$$K(x) = (\sqrt{y})^2 = y.$$

$\square$

Note:

K is also injective; hence K is bijective.

Ex. 7.2.17. The function f is defined on a set of real numbers. Determine whether or not f is one-to-one and justify your answer.

$$f(x) = \frac{3x-1}{x}, \text{ for all real numbers } x \neq 0.$$

One-to-one functions.

Definition

Let F be a function from a set X to a set Y. F is **one-to-one** (or **injective**) if and only if for all elements x_1 and x_2 in X,

$$\text{if } F(x_1) = F(x_2), \text{ then } x_1 = x_2.$$

Ex. 7.2.36, 37. If $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ are functions, then the function $(f+g) : \mathbb{R} \to \mathbb{R}$ is defined by the formula $(f+g)(x) = f(x) + g(x)$ for all real numbers x.

36. If f and g are both one-to-one, is $f+g$ also one-to-one? Justify your answer.

37. If f and g are both onto, is $f+g$ also onto? Justify your answer.

SOLUTION . **36.** No, $f+g$ need not be one-to-one.
Counterexample:

Take $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x$, and $g : \mathbb{R} \to \mathbb{R}$, $g(x) = -x$. It is clear that both f and g are one-to-one. Then for any $x \in \mathbb{R}$,

$$(f+g)(x) = f(x) + g(x) = x + (-x) = x - x = 0.$$

Take $x_1 = 1$ and $x_2 = 2$. Clearly $x_1 \neq x_2$ but $(f+g)(1) = 0$ and $(f+g)(2) = 0$ and so $(f+g)(1) = (f+g)(2)$.

37. No, $f+g$ need not be onto.
Counterexample:

Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x$ and $g(x) = -x$ just like the counterexample in the previous question.

From previous question, for all $x \in \mathbb{R}$, $(f+g)(x) = 0$. And so $f+g$ is not onto. (In particular, we see that $y = 1 \in \mathbb{R}$ does not have a pre-image. $\square$

Ex. 7.3.18. If $f : X \to Y$ and $g : Y \to Z$ are functions and $g \circ f$ is one-to-one, must f be one-to-one? Prove or give a counterexample.

SOLUTION . Yes, f must be one-to-one. Suppose $f : X \to Y$ and $g : Y \to Z$ are functions such that $g \circ f$ is one-to-one.

Let x_1 and x_2 be two elements of X such that $f(x_1) = f(x_2)$. Clearly $f(x_1) = f(x_2) \in Y$, and so

$$g(f(x_1)) = g(f(x_2)).$$

Therefore,

$$g \circ f(x_1) = g \circ f(x_2).$$

Since $g \circ f$ is one-to-one, we must have $x_1 = x_2$. This shows that f is one-to-one. $\square$

Ex. 7.3.19. If $f : X \to Y$ and $g : Y \to Z$ are functions and $g \circ f$ is onto, must g be onto? Prove or give a counterexample.

SOLUTION . Yes, g must be onto. Suppose $f : X \to Y$ and $g : Y \to Z$ are functions such that $g \circ f$ is onto. Let z be an element in Z. Since $g \circ f$ is onto, there exists $x \in X$ such that

$$g \circ f(x) = g(f(x)) = z.$$

Let $y = f(x)$. Clearly then $y \in Y$ and $g(y) = z$. Therefore, there exists an element y in Y such that $g(y) = z$. This shows that g is onto. $\square$

Ex. 7.3.24. Define $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ by the formulas

$$f(x) = x + 3 \qquad \text{and} \qquad g(x) = -x \qquad \text{for all } x \in \mathbb{R}.$$

Find $g \circ f$, $(g \circ f)^{-1}, g^{-1}, f^{-1}$, and $f^{-1} \circ g^{-1}$, and state how $(g \circ f)^{-1}$ and $f^{-1} \circ g^{-1}$ are related.

SOLUTION . It is clear that f and g are one-to-one functions and both f and g are onto functions. Therefore, their inverses exist. By theorem in lecture, we know that $g \circ f$ is also one-to-one and onto, and so the inverse of $g \circ f$ exists.

- The composition function $g \circ f : \mathbb{R} \to \mathbb{R}$ is defined by
$$g \circ f(x) = g(x+3) = -(x+3) = -x - 3 \qquad \text{for all } x \in \mathbb{R}.$$
- Let $x \in \mathbb{R}$. Suppose for some $z \in \mathbb{R}$, we have $z = g(f(x))$. Then $z = -x - 3$, i.e. $x = -z - 3$. Therefore, $(g \circ f)^{-1} : \mathbb{R} \to \mathbb{R}$ is defined by $(g \circ f)^{-1}(z) = -z - 3$ for all $z \in \mathbb{R}$.
- Let $y \in \mathbb{R}$. Suppose for some $z \in \mathbb{R}$, we have $z = g(y)$. Then $z = -y$. Then $y = -z$. Therefore, $g^{-1} : \mathbb{R} \to \mathbb{R}$ is defined by $g^{-1}(z) = -z$ for all $z \in \mathbb{R}$.
- Let $x \in \mathbb{R}$. Suppose for some $y \in \mathbb{R}$, we have $y = f(x)$. Then $y = x + 3$. Then $x = y - 3$. Therefore, $f^{-1} : \mathbb{R} \to \mathbb{R}$ is defined by $f^{-1}(y) = y - 3$ for all $y \in \mathbb{R}$.
- The composition function $f^{-1} \circ g^{-1} : \mathbb{R} \to \mathbb{R}$ is defined by
$$f^{-1}(g^{-1}(z)) = f^{-1}(-z) = -z - 3 \qquad \text{for all } z \in \mathbb{R}.$$

By the above and the definition of equality of functions, $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. $\square$

Bonus Qns: AY23/24 Finals

(b) Write down three functions $f_0 : \mathbb{Z} \to \mathbb{Z}$, $f_1 : \mathbb{Z} \to \mathbb{Z}$ and $f_2 : \mathbb{Z} \to \mathbb{Z}$ such that:

(i) f_0 is one-to-one but not onto.

(ii) f_1 is onto but not one-to-one.

(iii) f_2 is neither one-to-one nor onto.

Justify your answers.

(c) Suppose that $g : A \to B$ is a function. Prove that if $C \subseteq B$ and $D \subseteq B$, then

$$g^{-1}(C \cup D) = g^{-1}(C) \cup g^{-1}(D).$$

https://youtu.be/Uj3_Kqkl9Zo?si=0qKG2BlyshnvclUT